

Practice Problem
Logic

1. Consider a vocabulary with only four propositions, A, B, C, and D. How many models are there for the following sentences? Remember a model is a possible configuration of the world where the sentence is true. Also mention whether the sentences are valid, satisfiable or unsatisfiable.
 - a. $B \vee C$.
 - b. $\neg A \vee \neg B \vee \neg C \vee \neg D$.
 - c. $(A \Rightarrow B) \wedge A \wedge \neg B \wedge C \wedge D$.

Solution:

A	B	C	D	$B \vee C$	$\neg A \vee \neg B \vee \neg C \vee \neg D$	$A \Rightarrow B$	$A \wedge \neg B \wedge C \wedge D$	$(A \Rightarrow B) \wedge A \wedge \neg B \wedge C \wedge D$
F	F	F	F	F	T	T	F	F
F	F	F	T	F	T	T	F	F
F	F	T	F	T	T	T	F	F
F	F	T	T	T	T	T	F	F
F	T	F	F	T	T	T	F	F
F	T	F	T	T	T	T	F	F
F	T	T	F	T	T	T	F	F
F	T	T	T	T	T	T	F	F
T	F	F	F	F	T	F	F	F
T	F	F	T	F	T	F	F	F
T	F	T	F	T	T	F	F	F
T	F	T	T	T	T	F	T	F
T	T	F	F	T	T	T	F	F
T	T	F	T	T	T	T	F	F
T	T	T	F	T	T	T	F	F
T	T	T	T	T	F	T	F	F

- a) 12, satisfiable
- b) 15, satisfiable
- c) 0, unsatisfiable

2. Prove (by enumerating the truth values where applicable) each of the following assertions
 - a. α is valid if and only if $\text{True} \models \alpha$.
 - b. For any α , $\text{False} \models \alpha$.
 - c. $\alpha \models \beta$ if and only if the sentence $(\alpha \Rightarrow \beta)$ is valid.
 - d. $\alpha \equiv \beta$ if and only if the sentence $(\alpha \Leftrightarrow \beta)$ is valid.
 - e. $\alpha \models \beta$ if and only if the sentence $(\alpha \wedge \neg \beta)$ is unsatisfiable.

Solution:

- a) By definition **True** $\models \alpha$ is always true when α is True
- b) **False** $\models \alpha$ is always true irrespective of whether α is true or false.

α	β	$\alpha \Rightarrow \beta$
F	F	T
F	T	T
T	F	F
T	T	T

- c) if $\alpha \Rightarrow \beta$ is valid, then all the rows except row 3 are in the model. For these three rows, if α is true then β is also true, so $\alpha \models \beta$

α	β	$\alpha \Leftrightarrow \beta$
F	F	T
F	T	F
T	F	F
T	T	T

- d) From the truth table above α is logically equivalent to β when $\alpha \Leftrightarrow \beta$ is valid

α	β	$\alpha \wedge \neg \beta$
F	F	F
F	T	F
T	F	T
T	T	F

- e) According to the definition, a sentence is satisfiable if it is true in some model and unsatisfiable if it is not true in any single model. If assumption is made that $\alpha \wedge \neg \beta$ is unsatisfiable (all rows except row 3 in the truth table above) then it can be concluded that $\alpha \models \beta$, because β is true if α is true in the three rows above

3. Prove by enumerating the truth values that $(A \Leftrightarrow B) \wedge (\neg A \vee B)$ is satisfiable.

Solution:

The sentence $(A \Leftrightarrow B) \wedge (\neg A \vee B)$ is true in some models, so it is satisfiable

A	B	$A \vee B$	$\neg(A \Rightarrow B)$	$(A \vee B) \wedge \neg(A \Rightarrow B)$	$(A \Leftrightarrow B)$	$(\neg A \vee B)$	$(A \Leftrightarrow B) \wedge (\neg A \vee B)$
F	F	F	F	F	T	T	T
F	T	T	F	F	F	T	F
T	F	T	T	T	F	F	F
T	T	T	F	F	T	T	T

4. Give a KB = $\{A, B, A \Leftrightarrow C, B \wedge C \Rightarrow \neg(D \vee E)\}$, Prove $KB \models \neg D$. Show your work.

Solution

1. A
2. B
3. $A \Leftrightarrow C$
4. $B \wedge C \Rightarrow \neg(D \vee E)$
5. $(A \Rightarrow C) \wedge (C \Rightarrow A)$ [Biconditional Elimination on 3]
6. $(A \Rightarrow C)$ [And elimination on 5]
7. C [Modes Ponens on 1 and 6]
8. $\neg(D \vee E)$ [Modes Ponens on 2, 7 and 4]
9. $\neg D \wedge \neg E$ [De Morgan on 8]
10. $\neg D$ [And elimination on 9]

5. Give a $KB = \{\neg A, B, (C \vee D) \Rightarrow A\}$, Prove $KB \models \neg D$ by using logical equivalence, Modes Ponens and And elimination rules. Show your work.

Solution

1. $\neg A$
2. B
3. $(C \vee D) \Rightarrow A$
4. $\neg A \Rightarrow \neg(C \vee D)$ [contraposition on 3]
5. $\neg(C \vee D)$ [Modes Ponens on 1 and 4]
6. $\neg C \wedge \neg D$ [De Morgans on 5]
7. $\neg D$ [And Elimination on 6]